Input: $v \in \mathcal{N}(T)$, $p \in \{0, \star, ?\}^n$

Output: a list of indices J and a leaf ℓ

Initialize $J \leftarrow []$, $u \leftarrow v$, $p' \leftarrow p$
while u is not a leaf **do**
    $D^{v,p'} \leftarrow \{j \in [n] \colon p'_j = ? \wedge \operatorname{rank}(Q^{\preceq u}_{S(p',\star)+j}) = \operatorname{rank}(Q^{\preceq u}_{S(p',\star)}) + 1\}$
    for $j \in D^{v,p}$ **do** ▷ in arbitrary order
        $p'_j \leftarrow 0$
        $w_j \leftarrow u$
        $J \leftarrow [J, j]$
    end for
    Infer $b^u \leftarrow \langle x, Q^u \rangle$ assuming $(Q^{\prec v})^T x = b^{\prec v}$ and $x_j = 0$ for all $j \in S(p', 0)$
    Move $u \leftarrow \text{child}(u, b^u)$
end while
return (J, u)